Preference-Aware Coefficient Correction for Rewarded-Soups-Style Model Merging

Motivation and Goal

$$\theta_i \to \delta_i = \theta_i - \theta_0 \to R \to \lambda^* = f(p, R)$$

Motivation:

User preferences p are defined over objectives, but model merging happens in task-vector space.

Goal:

Use task-vector geometry R to compute preference-aware merge coefficients λ*.

Architecture Overview + Method

- δ ∈ R^P

$$D = [\delta_1, \ldots, \delta_m]$$

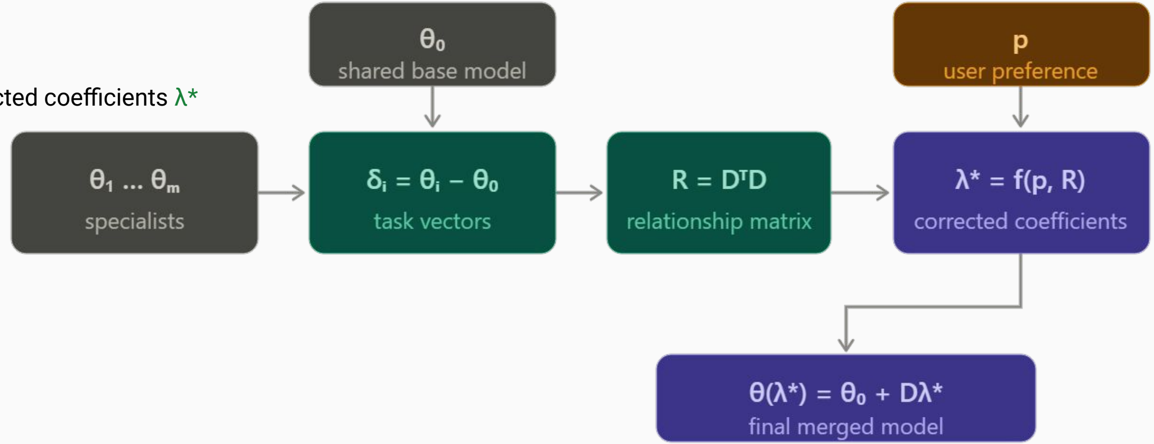
- D ∈ R ^ (P x m)
- Base model: TinyLlama-1.1B
- Five HelpSteer2 LoRA specialists
- Compute relationship matrix:
- R ∈ R ^(m x m)
- Compute corrected coefficients:
- λ ∈ R ^ m
- Merge adapters and evaluate with ArmoRM

$$R = D^\top D$$

$$\lambda^* = f(p, R)$$

1. Train attribute adapters
2. Extract task vectors
3. Compute relationship matrix R
4. Given preference vector p, solve for corrected coefficients λ*
5. Merge adapters
6. Evaluate merged model

$$U_p(\theta) := \sum_{i=1}^{m} p_i\, r_i(\theta)$$



Methods

$$\textbf{Avg}: \quad \lambda_{\text{Avg}} = \arg\max_{\lambda \in \Delta_m} \; p^\top R\lambda - \rho\,(\lambda - p)^\top R(\lambda - p),$$

$$\textbf{Cert}: \quad \lambda_{\text{Cert}} = \arg\max_{\lambda \in \Delta_m} \; \min_i \Delta_i(\lambda) \quad \text{s.t.} \quad (\lambda - p)^\top R(\lambda - p) \le c^2 \max(p^\top Rp, \varepsilon),$$

$$\textbf{MaxMin}(c): \quad d^\star = \arg\max_{\mathbf{1}^\top d = 0} \; \min_i (Rd)_i \quad \text{s.t.} \quad \|d - d_0\|_R \le c\,\|d_0\|_R\,, \quad d_0 = \Pi_0(Rp),$$

$$\lambda_{\text{MaxMin}(c)} = p + d^\star,$$

$$\textbf{Fair}(\alpha, \varepsilon): \quad \lambda_{\text{Fair}} = \arg\max_{\lambda \in \mathcal{F}_p^\varepsilon} \; U_\alpha\big(\Delta(\lambda) + \varepsilon\mathbf{1}\big) \quad \text{s.t.} \quad (\lambda - p)^\top R(\lambda - p) \le c^2,$$

RS-PPO-Training

- PPO is run once per reward/objective axis
- Hyperparameters are intended to stay fixed across axes
- Only the reward head changes between axes
- ArmoRM is used for both training and evaluation

RS-PPO-Training

- Restructured the RS-PPO pipeline
- Training similar to Rewarded Soups Paper
- ~11 hours per adapter
- Difficulties implementing

Geometry Check

- The adapters are not strongly conflicting
- They are mildly aligned with each other
- No clearly negative adapter relationships were found

Thesis Text

- main.tex was substantially revised

Next steps

- Test the proposed coefficient methods on the trained RS-PPO adapters
- Run the pipeline on another dataset to check whether the findings generalize
- Continue work on main.tex